

D859

General Geology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A field-trip leader, who has guided groups at a canyon for 20 years, points to tilted, eroded sedimentary layers covered by flat-lying younger sandstone. The canyon's visitor center sells detailed trail guides. What is this contact?

- A. A disconformity.
- B. An angular unconformity.
- C. A nonconformity.
- D. A fault, since all tilted layers have been faulted.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Flat-lying sandstone rests directly on an eroded surface of granite. What type of unconformity is this?

- A. An angular unconformity.
- B. A disconformity.
- C. None; the granite is younger than the sandstone above it.
- D. A nonconformity.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A petroleum company with offices in three countries is matching rock layers between two wells 60 km apart. Layer X contains an ammonite species that existed for only about 1 million years but lived in oceans worldwide. Layer Y contains horseshoe crabs, a group that has lived for hundreds of millions of years. The wells were drilled last spring. Which layer is more useful for matching ages between the wells?

- A. Either one, since any fossil works equally well.
- B. Layer Y, because long-lived fossils appear in more layers.
- C. Layer X, because its fossil lived for a short time over a wide area.
- D. Layer Y, because horseshoe crabs are still alive and easy to study.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Scientists at a coastal university, which runs a popular summer field camp, measure land along a northern bay rising about 1 cm per year, although no plates converge there. The region was covered by a thick ice sheet until about 10,000 years ago. What explains the uplift?

- A. Magma is pushing the crust up from below.
- B. The land is still sinking under the ice's weight.
- C. Isostatic rebound: crust pushed down by the ice is slowly rising as mantle rock flows back.
- D. Sea level is falling everywhere in the world.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A research ship with a crew of 40 tows a magnetometer across a mid-ocean ridge and finds stripes of normal and reversed magnetism in the seafloor. The ship also collects deep-sea mud samples. Which pattern supports seafloor spreading?

- A. The stripes are bands of different minerals.
- B. The stripes are mirror images on both sides of the ridge, and the rock gets older away from the ridge.
- C. The youngest rock is farthest from the ridge.
- D. The stripes appear on only one side of the ridge.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A basalt sample is taken 300 km from a mid-ocean ridge. Compared with basalt at the ridge crest, it is:

- A. Older.
- B. Younger.
- C. The same age, since the whole seafloor formed at once.
- D. Older, and it formed at a subduction zone.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A hotspot island chain, popular with snorkelers, has an extinct volcano now 3,000 km from the active hotspot; its lava is dated at 60 million years. The volcano's summit reaches 1,200 m. What was the average plate speed?

- A. 0.02 cm per year
- B. 2 cm per year
- C. 0.5 cm per year
- D. 5 cm per year

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A plate moves 4 cm per year. How long does it take to move 200 km?

- A. 5 million years
- B. 50 years
- C. 800 years
- D. 500,000 years

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

On a topographic map used by a county search-and-rescue team, contour lines are spaced far apart on the east side of a hill and packed close together on the west side. The contour interval is 20 m, and the map was printed in 2019. Which side is steeper?

- A. Neither, since the contour interval is the same on both sides.
- B. The east side.
- C. The west side.
- D. The west side is lower, not steeper.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Where contour lines cross a stream on a topographic map, they bend into V shapes. Which way do the V's point?

- A. Downstream.
- B. Upstream, toward higher ground.
- C. V's show a ridge, not a stream.
- D. Toward the highest peak on the map.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Hikers on a guided trip sponsored by an outdoor gear store cross a valley with a broad U-shaped cross section, steep walls, and side valleys that end in high waterfalls. Large granite boulders rest on limestone bedrock. The trip lasted six days. What best explains this landscape?

- A.** A valley glacier carved the valley and left the boulders as glacial erratics.
- B.** A river carved the valley over time.
- C.** The granite boulders weathered out of the limestone bedrock.
- D.** Earthquake faulting dropped the valley floor.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A museum preparing a new deep-time exhibit, funded by a local foundation, plans a panel on the thin clay layer that marks the end of the Cretaceous. The clay is rich in iridium, and non-bird dinosaur fossils do not appear above it. The panel will be 3 m wide. What does the evidence support?

- A.** The layer marks the start of the Mesozoic Era.
- B.** Iridium can come only from volcanoes, so no impact occurred.
- C.** Dinosaurs slowly evolved into mammals across the layer.
- D.** A mass extinction about 66 million years ago, linked to a large asteroid impact, marking the end of the Mesozoic Era.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A field-trip leader, ~~who has guided groups at a canyon for 20 years~~, points to tilted, eroded sedimentary layers covered by flat-lying younger sandstone. ~~The canyon's visitor center sells detailed trail guides.~~ What is this contact?

Target word: type of contact **Decides:** Tilted layers below, flat layers above.

A	Adjacent category	A disconformity lies between parallel layers.
B	Correct	Tilted, eroded layers beneath flat layers define an angular unconformity.
C	Confusion set	A nonconformity has sediment on igneous or metamorphic rock.
D	Absolute language	Layers can tilt by folding; the erosion surface marks an unconformity.

2. Answer D Concept 1 • transfer

Flat-lying sandstone rests directly on an eroded surface of granite. What type of unconformity is this?

Target word: unconformity type **Decides:** Sediment on igneous rock.

A	Confusion set	That needs tilted sedimentary layers below.
B	Adjacent category	That needs sedimentary layers on both sides.
C	Inverted operation	The granite was eroded before the sand was laid on it, so it is older.
D	Correct	Sedimentary rock over eroded igneous or metamorphic rock.

3. Answer C Concept 2

A petroleum company ~~with offices in three countries~~ is matching rock layers between two wells 60 km apart. Layer X contains an ammonite species that existed for only about 1 million years but lived in oceans worldwide. Layer Y contains horseshoe crabs, a group that has lived for hundreds of millions of years. ~~The wells were drilled last spring.~~ Which layer is more useful for matching ages between the wells?

Target word: best fossil for correlation **Decides:** Short time range, wide spread.

A	Dropped element	Drops the short-range, wide-spread criteria.
B	Inverted operation	Appearing in many layers is exactly what makes a fossil a poor time marker.
C	Correct	A short range pins the age; a wide range lets it be found in many places.
D	Adjacent category	Being easy to study is not what makes a fossil a good age marker.

4. Answer C Concept 3

Scientists at a coastal university, ~~which runs a popular summer field camp~~, measure land along a northern bay rising about 1 cm per year, although no plates converge there. The region was covered by a thick ice sheet until about 10,000 years ago. What explains the uplift?

Target word: cause of uplift **Decides:** Ice was removed from depressed crust.

A	Confusion set	No volcanic activity is described; the pattern matches glacial rebound.
B	Inverted operation	The ice is gone; the land is rising, not sinking.
C	Correct	Removing weight lets the floating crust rise.
D	Adjacent category	Global sea level has been rising; the change is in the land.

5. Answer B Concept 4

A research ship ~~with a crew of 40~~ tows a magnetometer across a mid-ocean ridge and finds stripes of normal and reversed magnetism in the seafloor. ~~The ship also collects deep-sea mud samples.~~ Which pattern supports seafloor spreading?

Target word: supporting pattern **Decides:** Symmetry about the ridge.

A	Confusion set	The stripes record reversals of Earth's magnetic field, not changes in mineral type.
B	Correct	New crust forms at the ridge and moves outward on both sides.
C	Inverted operation	The youngest rock is at the ridge crest.
D	Dropped element	Spreading goes both ways, so stripes appear on both sides.

6. Answer A Concept 4 • transfer

A basalt sample is taken 300 km from a mid-ocean ridge. Compared with basalt at the ridge crest, it is:

Target word: relative age from distance **Decides:** Age increases away from the ridge.

A	Correct	It formed at the ridge and has since moved 300 km away.
B	Inverted operation	Reverses the age pattern.
C	Absolute language	Seafloor forms continuously at ridges.
D	Confusion set	Right age, wrong origin; seafloor forms at ridges and is destroyed at subduction zones.

7. Answer D Concept 5

A hotspot island chain, ~~popular with snorkelers~~, has an extinct volcano now 3,000 km from the active hotspot; its lava is dated at 60 million years. ~~The volcano's summit reaches 1,200 m.~~ What was the average plate speed?

Target word: plate speed **Decides:** $3,000 \text{ km} \div 60 \text{ million years}$.

A	Inverted operation	Divides 60 by 3,000.
B	Camouflage	Uses the 1,200 m summit height as if it were a distance traveled.
C	Dropped element	Drops a factor of 10 in the unit conversion.
D	Correct	$3,000 \text{ km} \div 60 \text{ My} = 50 \text{ km/My}$; $50 \text{ km} = 5,000,000 \text{ cm}$, $\div 1,000,000 \text{ yr} = 5 \text{ cm/yr}$.

8. Answer A Concept 5 • transfer

A plate moves 4 cm per year. How long does it take to move 200 km?

Target word: time from rate **Decides:** $200 \text{ km} = 20,000,000 \text{ cm}$.

A	Correct	$20,000,000 \text{ cm} \div 4 \text{ cm/yr} = 5,000,000 \text{ years}$.
B	Dropped element	Divides 200 by 4 without converting km to cm.
C	Inverted operation	Multiplies 200×4 instead of dividing.
D	Wrong stem number	Loses a zero in the conversion.

9. Answer C Concept 6

On a topographic map ~~used by a county search and rescue team~~, contour lines are spaced far apart on the east side of a hill and packed close together on the west side. The contour interval is 20 m, and ~~the map was printed in 2019~~. Which side is steeper?

Target word: steeper side **Decides:** Contour spacing.

A	Wrong stem number	The interval is fixed for the whole map; spacing shows steepness.
B	Inverted operation	Wide spacing means a gentle slope.
C	Correct	Close contours mean elevation changes quickly over a short distance.
D	Confusion set	Spacing shows slope, not elevation.

10. Answer B Concept 6 · transfer

Where contour lines cross a stream on a topographic map, they bend into V shapes. Which way do the V's point?

Target word: direction of contour V's **Decides:** Rule of V's.

A	Inverted operation	Reverses the rule of V's.
B	Correct	Contours bend up the valley before crossing the stream.
C	Confusion set	V's pointing downhill mark ridges; those crossing a stream point upstream.
D	Adjacent category	They point up the local valley, not to a map-wide high point.

11. Answer A Concept 7

Hikers on a guided trip ~~sponsored by an outdoor gear store~~ cross a valley with a broad U-shaped cross section, steep walls, and side valleys that end in high waterfalls. Large granite boulders rest on limestone bedrock. ~~The trip lasted six days~~. What best explains this landscape?

Target word: origin of the landscape **Decides:** U-shape, hanging valleys, out-of-place boulders.

A	Correct	U-shaped valleys and erratics are glacial signatures.
B	Confusion set	Rivers carve V-shaped valleys.
C	Inverted operation	Granite cannot come from limestone; the boulders were carried in.
D	Adjacent category	Faulting does not produce U-shapes, hanging valleys, and erratics together.

12. Answer D Concept 8

A museum preparing a new deep-time exhibit, ~~funded by a local foundation~~, plans a panel on the thin clay layer that marks the end of the Cretaceous. The clay is rich in iridium, and non-bird dinosaur fossils do not appear above it. ~~The panel will be 3 m wide~~. What does the evidence support?

Target word: meaning of the boundary layer **Decides:** Iridium spike plus sudden fossil change.

A	Inverted operation	It marks the end of the Mesozoic and the start of the Cenozoic.
B	Absolute language	Iridium is concentrated in asteroids; "only" is false.
C	Confusion set	Mammals and dinosaurs are separate lineages; the change is abrupt, not gradual.
D	Correct	Iridium is rare in crust but common in asteroids; the fossil change is abrupt.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Types of unconformities	An angular unconformity has tilted layers under flat ones; a nonconformity has sedimentary rock on eroded igneous or metamorphic rock; a disconformity is an erosion surface between parallel sedimentary layers.	Mixing up the three types, or calling any tilted contact a fault.	A description of what lies above and below an erosion surface.
2	Index fossils	The best fossils for matching ages between places lived for a short time but over a wide area.	Choosing fossils that are long-lived or abundant.	Two fossils compared by time range and geographic spread.
3	Isostasy	The crust floats on the denser mantle; adding weight (ice, sediment) pushes it down, and removing weight lets it slowly rise (rebound).	Explaining uplift with plate collision or global sea-level fall.	Uplift in a formerly glaciated area with no plate collision.
4	Seafloor spreading evidence	Magnetic stripes are mirror images on either side of a mid-ocean ridge, and seafloor gets older with distance from the ridge.	Thinking the oldest seafloor is at the ridge.	Rocks sampled at different distances from a ridge.
5	Plate speed from distance and age	Rate = distance ÷ time; 1 km per million years = 0.1 cm per year.	Dividing the wrong way or losing a factor of 10 in unit conversion.	A distance and an age to be turned into cm per year.
6	Reading contour lines	Close contour lines mean steep slopes; wide spacing means gentle slopes; contours form V's that point upstream where they cross a stream.	Thinking close lines mean high ground or that V's point downstream.	A map with uneven contour spacing or contours crossing a stream.
7	Glacial landforms	Valley glaciers carve U-shaped valleys with hanging valleys and leave erratics — boulders carried far from their source rock; rivers carve V-shaped valleys.	Crediting a U-shaped valley to a river.	A U-shaped cross section and out-of-place boulders.
8	Mass extinctions mark era boundaries	Major boundaries in the geologic time scale are set by mass extinctions; the end-Cretaceous (about 66 million years ago) extinction, marked by an iridium-rich clay layer, is linked to an asteroid impact.	Reading the boundary backward or explaining it as slow evolution.	A thin boundary layer with a sharp change in fossils.

Course scope (public description): General Geology offers a comprehensive exploration of Earth's dynamic systems, emphasizing the interaction of geological processes that shape our planet. Students will engage in a rigorous study of Earth's materials, surface processes, and the forces driving tectonic movements, with an emphasis on both local and global geological phenomena. Through interactive activities, including simulations, and mapping tools, learners will connect historical geological evidence to the evolution of the Earth and its life forms.

The course is designed for educators aiming to enhance their pedagogical approaches to teaching geology. No prior geological knowledge is required, making this course accessible and essential for anyone interested in understanding the Earth's past, present, and future.